

1 Chapter 1

Nick's Question 1.1 Proof of lemma 1.8.4

The proof is very nice.

2 Chapter 2

Nick's Question 2.1 In the proof of Theorem 2.4.3 on the bottom of page 34 there are some claims made that I am unable to verify why they are true. First, we are considering

$$X_t = X_0 + \int_0^t u(s)dB_s$$

and here $u(s)$ only satisfies

$$\mathbb{P}\left(\int_0^t u(s)^2 ds < \infty\right) = 1$$

so, as in the Kuo text, this stochastic integral exists and a limit in probability of a sequence of stochastic integrals $\int_0^t u_n(s)dB_s$ where the u_n are simple and are in L^2_{ad} and hence $\int_0^t u_n(s)dB_s$ is a martingale and we know the first and second moment. But for the limit $\int_0^t u(s)dB_s$ we know nothing about the moments. But in this text, it is claimed that

$$\mathbb{E}\left[\int_{t_i}^{t_{i+1}} u^2(s)ds - \left(\int_{t_i}^{t_{i+1}} u(s)dB_s\right)^2 \middle| \mathcal{F}_i\right] = 0$$

and I have never seen this claimed in any other text.

Note that u is bounded (we are assuming localization has been done). Hence, all moments exists.

Nick's Question 2.2 In the proof of theorem 2.5.1, why is it true that

$$E\left(\int_0^t |u'_n(B_s) - 1_{(x,\infty)}(B_s)|^2 ds\right) \leq \int_0^t \mathbb{P}\left(|B_s - x| \leq \frac{2}{n}\right) ds$$

Please explore the mollifier f and f_n . Plot these u'_n and see how could it differ from $1_{(x,\infty)}$.

Nick's Question 2.3 What is the motivation on page 41 behind the following definition of the backward ito integral

$$\int_0^T u(s)\hat{d}B_s = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-2} \left(\frac{n}{T} \int_{(j+1)T/n}^{(j+2)T/n} u(s)ds\right) (B_{(j+1)T/n} - B_{jT/n})$$

The integral in the parenthesis is the average of u_s on $((j+1)T/n, (j+2)T/n)$.

Let's skip this part. I know nothing on the backward SDE and not particular interested either. It has applications in finance. If you don't do finance, I don't see the point here.

Nick's Question 2.4 (Problem 2.7)

This problem follows easily from Levy's Characterization of Brownian Motion. But I can't figure out how to do this problem with the hint given.

This is a nice problem. The hints is to ask you to find the distribution of $X_t - X_s$ through the characteristic function. Once apply Ito's formula, take expectation and remove the martingale part, you should arrive at an integral equation. Take derivatives, you arrive at a differential equation. The solution for $a_s(t) := \mathbb{E}(e^{i\lambda(X_t - X_s)})$ is

$$a_s(t) = e^{-\frac{\lambda^2}{2}(t-s)}.$$

Nick's Question 2.5 (Problem 2.15)

First we consider the 2 - d Ito process

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} E(X^p) \\ E(Y^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}$$

and then following the hint we apply Ito's formula to $f(x, y) = x^{1/p}y^{1/q'}$. We get the following derivatives:

- $f_x = \frac{1}{p}x^{\frac{1}{p}-1}y^{\frac{1}{q'}}$
- $f_{xx} = \frac{1}{p}\left(\frac{1}{p}-1\right)x^{\frac{1}{p}-2}y^{\frac{1}{q'}}$
- $f_y = \frac{1}{q'}x^{\frac{1}{p}}y^{\frac{1}{q'}-1}$
- $f_{yy} = \frac{1}{q'}\left(\frac{1}{q'}-1\right)x^{\frac{1}{p}}y^{\frac{1}{q'}-2}$

Now putting this into Ito's Formula gives us

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p}-1\right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'}-1\right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \end{aligned}$$

I am not really sure where to continue from here. I see that if we let $t \rightarrow \infty$ then $M_t^{1/p} N_t^{1/q'} \rightarrow XY$ so if we took the expected value of both sides then we would partially have what we are looking for since $E(XY)$ is on the left and $\|X\|_p \|Y\|_{q'}$ is on the right. But I am not sure about the other terms. Im guessing the Ito integrals in the right have 0 expectation but I am not sure if that is in fact true since I dont see how it is easily seen that the integrands are in $L^2(\mathcal{P})$.

What I did next was see what $\hat{B}_s$ was equal too and for this we define another Ito Process

$$\begin{pmatrix} B_s \\ \tilde{B}_s \end{pmatrix} = \int_0^t \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} dB_s \\ d\tilde{B}_s \end{pmatrix}$$

and applied Itos formula to $f(t, x, y) = e^{-t}x + \sqrt{1 - e^{-2t}}y$ which gives us

$$\hat{B}_s = e^{-t}B_s + \sqrt{1 - e^{-2t}}\tilde{B}_s = \int_0^t \frac{e^{-2s}}{\sqrt{1 - e^{-2s}}}\tilde{B}_s - e^{-s}B_s ds + \int_0^t e^{-s}dB_s + \int_0^t \sqrt{1 - e^{-2s}}d\tilde{B}_s$$

However I am not sure if this step is necessary.

The followings are some of my thoughts: First of all, we need the absolute values:

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} \mathbb{E}(|X|^p) \\ \mathbb{E}(|Y|^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}.$$

Second, you forget

$$f_{xy} = \frac{1}{p} \frac{1}{q'} x^{-1/(p')} y^{-1/q'}.$$

Hence, by Ito's formula, we will have

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p} - 1 \right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'} - 1 \right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \\ &\quad + \frac{1}{pq'} \int_0^t M_s^{-\frac{1}{p'}} N_s^{-\frac{1}{q}} \psi_s \phi_s d\langle B, \hat{B} \rangle_s \\ &=: \|X\|_p \|Y\|_{q'} + \sum_{i=1}^5 I_i. \end{aligned}$$

Upon taking expectation on both sides, we see that

$$LHS = \mathbb{E}(|X_t Y_t|) \geq \mathbb{E}(X_t Y_t).$$

We only need to prove that

$$\mathbb{E} \left(\sum_{i=1}^5 I_i \right) \leq 0,$$

which is clear for $i = 1, 2$, in which case we have the expectation equal to zero. Now for the rest three parts, we see that

$$I_3 + I_4 + I_5 = -\frac{1}{2pq'} M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}-2} \left(N_s^2 \varphi_s^2 \frac{q'}{p'} - 2M_s N_s \phi_s \psi_s e^{-s} + M_s \psi_s^2 \frac{p}{q} \right),$$

where we have used the fact that $d\langle B, \hat{B} \rangle_s = e^{-s} ds$. Hence, the condition become the condition that the quadratic form has no solution:

$$\Delta = 4e^{-2s} - 4\frac{q'p}{p'q} = 4 \left(e^{-2s} - \frac{p-1}{q-1} \right) \leq 0$$

which is equivalent to

$$q \leq e^{2s}(p-1) + 1.$$

Hence, the largest q that makes the inequality work is $q = e^{2s}(p-1) + 1$.

3 Chapter 3

3.1 Reading Material

Nick's Question 3.1 (Smooth Cylindrical Random Variable) Following chapter 3 of the Kuo text, denote C the Wiener space of continuous functions from $[0,1]$ to $\mathbb{R}$, $\mathcal{R}$ the collection cylindrical sets

$$A = \{\omega \in C : (\omega(t_1), \dots, \omega(t_n)) \in U \mid t_i \in [0, 1], n \in \mathbb{N}, U \in \mathcal{B}(\mathbb{R}^n)\}$$

and μ the associated Wiener measure. Consider the coordinate processes $B(t, \omega) = \omega(t)$ which are Brownian motions under $(C, \mathcal{B}(\mathcal{R}), \mu)$. There is a theorem that can be found [here](#).

Theorem : The random variables

$$\{f(B(t_1, \cdot), \dots, B(t_n, \cdot)) \mid n \in \mathbb{N}, f \in \mathcal{S}(\mathbb{R}^n)\}$$

are dense in $L^2(C, \mu)$ where $\mathcal{S}(\mathbb{R}^n)$ are the Schwartz functions.

How does this relate to $\mathcal{S}$ defined on page 51 of the intro to Malliavin textbook? How are the cylindrical random variables defined in this case? Why is $\mathcal{S}$ dense in $L^2(\Omega)$? Even in Diffusions Markov Processes and Martingales - Roger and Williams, they define the cylindrical sigma field in the same way Kuo does- through the use of coordinate processes.

Nick Solution.

density

Theorem 3.1. Consider the probability space $(\Omega, \mathcal{F}, P)$ where $\mathcal{F}$ is the standard Brownian filtration. Then $\{f(B(h_1), \dots, f(B(h_n)), n \in \mathbb{N}, f \in C_p^\infty(\mathbb{R}^n), h_i \in H := L^2(\mathbb{R}^+)\}$ is dense in $L^2(\Omega)$.

Proof: This proof follows the same scheme as the solution found [here](#) which utilizes Dynkin's multiplicative system theorem. First, we can easily show that the sigma field generated by $\{B(h)\}_{h \in H}$ generates $\mathcal{F}$. This is true since $1_{[0,t]} \in H$ and $B(1_{[0,t]}) = B_t$. Second, since $\mathcal{S}(\mathbb{R}) \subset C_p^\infty(\mathbb{R})$ and since for every $h \in H$ there exists functions $f_n \in \mathcal{S}(\mathbb{R})$ such that $f_n \rightarrow h$ in $L^2(\mathbb{R})$ then by the Wiener Isometry we have $E|\int_0^\infty h(s) - f_n(s)dB_s|^2 = \int_0^\infty |h(s) - f_n(s)|^2 ds \rightarrow 0$. This implies that the sigma field generated by $\{f(B(h))\}$ where $h \in H$ and $f \in C_p^\infty(\mathbb{R})$ is the same as $\mathcal{F}$.

Now we let $M = \{f(B(h_1), \dots, B(h_n))\}$. We have already shown that $\sigma(M)$ is the same as $\mathcal{F}$. Also, M is closed under pointwise multiplication. Let $E := \bar{M}$ where closure is with respect to the $L^2(\Omega)$ norm. Let $H := E \cap L^\infty(\Omega)$. We have $M \subset H$. It can be shown that H is closed with respect to bounded pointwise convergence and also that H contains all constant functions. Thus, the hypothesis is Dynkin's multiplicative theorem has been satisfied and this implies that H contains all $\sigma(M) = \mathcal{F}$ measurable functions. But this means that it contains all bounded Borel measurable functions. But by the definition of H , this implies that E contains all the bounded Borel measurable functions. We know that all $X \in L^2(\Omega)$ can be approximated in the $L^2(\Omega)$ norm by bounded measurable functions. Also since E is closed and contains the bounded measurable functions, then E itself must be equal to $L^2(\Omega)$. Therefore we conclude that $\bar{M} = E = L^2(\Omega)$. □

Nick's Question 3.2 Why is $(D_t F)_{t \geq 0}$ a stochastic processes in $L^2(\Omega; H)$ where $H = L^2(\mathbb{R}^+)$? $D_t F$ is defined as

$$D_t F = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(B(h_1), \dots, B(h_n)) h_i(t)$$

where $h_i \in H$ and $B(h_i) = \int_0^\infty h_i dB_s$ is the standard Wiener Integral. So isn't $D_t F$ real

valued since we are evaluating the $h_i(\cdot)$ at t ? Then shouldn't the $(D_t F)_{t \geq 0}$ be a $L^2(\Omega; \mathbb{R})$ valued stochastic process?

I believe that this is just a typo in the text

Nick's Question 3.3 (Proposition 3.2.4)(All questions have been solved with work below)

1. Why does $D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u)$? (I have answered this below)
2. Also, at the bottom of page 52, why is it that (I have answered this below)

$$E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle \right) = E \left(\int_0^{\infty} \int_0^{\infty} D_s u_t D_t v_s ds dt \right)$$

3. In the proof of property 3.3 why do we have (Answered below)

$$E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \left(\langle u, e_i \rangle + \delta(D_{e_i}(u)) \right) \right) = E(\langle u, v \rangle) + E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle e, e_j \rangle D_{e_j} \langle v, e_i \rangle \right)$$

Proof of $D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u)$.

Since $u = \sum F_j h_j$ we just apply the definition of δ to get

$$\begin{aligned} D_h(\delta(u)) &= D_h \left(\sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle \right) \\ &= \sum_{j=1}^n D_h(F_j B(h_j)) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\ &= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \end{aligned} \quad (3.1) \quad \boxed{\text{P:3.2.4-1}}$$

where the third inequality follows from the product rule and $D_h(B(h_j)) = \langle h, h_j \rangle$. Now we focus on the last term from the last line above (3.1).

$$\sum_{j=1}^n D_h \langle DF_j, h_j \rangle = \sum_{j=1}^n \langle D_t \langle DF_j, h_j \rangle, h(t) \rangle \quad (3.2) \quad \boxed{\text{P:3.2.4-6}}$$

Before we continue, we simplify $D_t \langle DF_j, h_j \rangle$. We may assume that $F_j = f^j(B(h_1^j(t)), \dots, B(h_n^j(t)))$ where $f^j \in C_p^\infty(\mathbb{R}^n)$ and $h_i^j \in H$.

$$\begin{aligned} D_t \langle DF_j, h_j \rangle &= D_t \left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(s), h_j(s) \right\rangle \\ &= \sum_{i=1}^n \langle h_i^j(s), h_j(s) \rangle D_t \left(\frac{\partial f^j}{\partial x_i} \right) \end{aligned} \quad (3.3) \quad \boxed{\text{P:3.2.4-2}}$$

We now plug (3.3) into (3.2) above to get

$$\sum_{j=1}^n \langle D_t \langle DF_j, h_j \rangle, h(t) \rangle = \sum_{j=1}^n \sum_{i=1}^n \langle h_i^j(s), h_j(s) \rangle \langle D_t \left(\frac{\partial f^j}{\partial x_i} \right), h(t) \rangle$$

$$\begin{aligned}
&= \sum_{j=1}^n \sum_{i=1}^n \left\langle h_i^j(s) \left\langle D_t \left(\frac{\partial f^j}{\partial x_i} \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_t \left(\sum_{i=1}^n h_i^j(s) \frac{\partial f^j}{\partial x_i} \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_t (DF_j), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_h (DF_j), h_j(s) \right\rangle
\end{aligned}$$

and if I haven't made a mistake, then what we have shown so far is that

$$\begin{aligned}
D_h(\delta(u)) &= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\
&= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \left\langle D_h (DF_j), h_j \right\rangle
\end{aligned} \tag{3.4} \quad \boxed{\text{P:3.2.4-3}}$$

Also, the above is exactly line two of the first centered equations on page 53.

Now we need to show that the last two terms add up to $\delta(D_h(u))$. This is not defined in the text but I think that we need to define $D_h u$ as the following

$$D_h u = D_h \sum_{j=1}^n F_j h_j(t) = \sum_{j=1}^n h_j(t) D_h(F_j) \tag{3.5} \quad \boxed{\text{P:3.2.4-4}}$$

Thus now $D_h(u)$ (3.5) is of the form $\sum F_j h_j$ and so we may apply the definition of δ to it to get the following

$$\delta(D_h u) = \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle \tag{3.6} \quad \boxed{\text{P:3.2.4-5}}$$

Now we focus in the last term of the right hand side of (3.6). From (3.4) we note that we want to show that

$$\sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle = \sum_{j=1}^n \left\langle D_h (DF_j), h_j \right\rangle \tag{3.7} \quad \boxed{\text{P:3.2.4-7}}$$

Here are the steps,

$$\begin{aligned}
\sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle &= \sum_{j=1}^n \left\langle D_s \left(\left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(t), h(t) \right\rangle \right), h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_s \left(\sum_{i=1}^n \frac{\partial f^j}{\partial x_i} \langle h_i^j(t), h(t) \rangle \right), h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} \right) \langle h_i^j(t), h(t) \rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} \right) h_i^j(t), h(t) \right\rangle, h_j(s) \right\rangle
\end{aligned}$$

$$\begin{aligned}
&= \sum_{j=1}^n \left\langle \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} h_i^j(t) \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_s(D_t F_j), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_h(D F_j), h_j \right\rangle
\end{aligned}$$

Hence (3.7) is shown and we are done.

Proof of bottom of page 52

$$\begin{aligned}
\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle &= \sum_{i,j=1}^{\infty} \left\langle D_s \langle u, e_j \rangle, e_i(s) \right\rangle \left\langle D_t \langle v, e_i \rangle, e_j(t) \right\rangle \\
&= \sum_{i,j=1}^{\infty} \left\langle \langle D_s(u_t), e_j(t) \rangle, e_i(s) \right\rangle \left\langle \langle D_t(v_s), e_i(s) \rangle, e_j(t) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \sum_{j=1}^{\infty} \left(\langle D_s(u_t), e_j(t) \rangle \left\langle \langle D_t(v_s), e_i(s) \rangle, e_j(t) \right\rangle \right), e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \left\langle D_s(u_t), \langle D_t(v_s), e_i(s) \rangle \right\rangle, e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \left\langle D_s(u_t), \langle D_t(v_q), e_i(q) \rangle \right\rangle, e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \int_0^{\infty} \left(\int_0^{\infty} D_s(u_t) \left(\int_0^{\infty} D_t(v_q) e_i(q) dq \right) dt \right) e_i(s) ds \\
&= \int_0^{\infty} \left(\int_0^{\infty} D_s(u_t) \left(\sum_{i=1}^{\infty} \int_0^{\infty} D_t(v_q) e_i(q) dq e_i(s) \right) dt \right) ds \\
&= \int_0^{\infty} \int_0^{\infty} D_s(u_t) D_t(v_s) dt ds
\end{aligned}$$

Where above we use the identity

$$\langle f, g \rangle = \sum_{i=1}^{\infty} \langle f, e_i \rangle \langle g, e_i \rangle.$$

Proof of step from property 3

$$\begin{aligned}
E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \left(\langle u, e_i \rangle + \delta(D_{e_i}(u)) \right) \right) &= E(\langle u, v \rangle) + E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \delta(D_{e_i}(u)) \right) \\
&= E(\langle u, v \rangle) + \sum_{i=1}^{\infty} E \left(\left\langle D(\langle v, e_i \rangle), D_{e_i}(u) \right\rangle \right) \quad (3.8) \quad \boxed{\mathbf{s-1}} \\
&= E(\langle u, v \rangle) + \sum_{i=1}^{\infty} E \left(\left\langle \langle D_s(v_t), e_i(t) \rangle, \langle D_t(u_s), e_i(t) \rangle \right\rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(\left\langle \langle D_s(v_t), e_i(t) \rangle, e_j(s) \right\rangle \left\langle \langle D_t(u_s), e_i(t) \rangle, e_j(s) \right\rangle \right)
\end{aligned}$$

$$\begin{aligned}
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v_t, e_i(t) \rangle \left\langle \langle D_t(u_s), e_j(s) \rangle, e_i(t) \right\rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v_t, e_i(t) \rangle D_{e_i} \langle u_s, e_j(s) \rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v, e_i \rangle D_{e_i} \langle u, e_j \rangle \right)
\end{aligned} \tag{3.9} \quad \boxed{\text{s-2}}$$

where in (3.8) we used Proposition 3.2.3 and in (3.9) we use Fubini's Theorem.

Nick's Question 3.4 (Why is δ linear?) On page 55 it says that δ is linear. First, for $u = \sum_{j=1}^n F_j h_j$ we have

$$\delta(u) = \sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle_H$$

Now if we have a $v = \sum_{j=1}^n G_j h'_j$ how does $\delta(u+v) = \delta(u) + \delta(v)$. Also how could we write $\sum_{j=1}^n F_j h_j + \sum_{j=1}^n G_j h'_j = \sum_{j=1}^n K_j k_j$ where $K_j \in \mathcal{S}$ and $k_j \in H$

Do we do this through the relationship $E(\langle DF, u \rangle) = E(\delta(u)F)$ in the sense that

$$\begin{aligned}
E(\delta(u+v)F) &= E(\langle DF, u+v \rangle) \\
&= E(\langle DF, u \rangle) + E(\langle DF, v \rangle) \\
&= E(\delta(u)F) + E(\delta(v)F) \\
&= E((\delta(u) + \delta(v))F)
\end{aligned}$$

and if this is true for every F then I believe we can say $\delta(u+v) = \delta(u) + \delta(v)$? **Yes, this is the argument to show that δ is linear. In general, for two distributions F and G , how one identifies $F = G$? This is true iff $\langle F, \varphi \rangle = \langle G, \varphi \rangle$ for all test function φ . Here, is the same. In order to identify that two random variables $\delta(u+v)$ and $\delta(v) + \delta(u)$ are same, this relation has to be tested against all functions in its dual space, which is $\mathbb{D}^{1,2}$.**

Nick's Question 3.5 Exercise 3.1 I don't see how we can apply this hint? Suppose we are able to show that the hint is true then how does that show $\mathcal{P}$ is dense in $L^q(\Omega)$? There is a theorem that $\{x_n\}$ is a basis for a Hilbert space H if and only if $\sum \langle x, x_n \rangle x_n = 0$ implies $x = 0$ but this is only valid for Hilbert space and $L^q(\Omega)$ for $q \neq 2$ is not a Hilbert space since there is no inner product. **This is the standard argument to show one set is dense in another. You may miss the fact that the set of polynomial functions is dense in $C_0^\infty(\mathbb{R}^d)$, the latter is dense in the set of the Schwartz test function $\mathcal{S}(\mathbb{R}^d)$. Please do justify yourself this problem.**

Nick's Question 3.6 Prop 3.3.2 page 54 This relates to exercise 3. In the proof of prop 3.3.2 how can we use Holders inequality to show that $\varphi'(F)D(F) \in L^q(\Omega, H)$?

Nicks Solution.

$$[E \|\varphi'(F)DF\|_H^q]^{1/q} = \left[\int_{\Omega} \left(\int_0^\infty \varphi'(F)^2 (D_t F)^2 dt \right)^{q/2} dP \right]^{1/q}$$

$$\begin{aligned}
&\leq \left[\int_{\Omega} (1 + |F|^{\alpha})^q \left(\int_0^{\infty} (D_t F)^2 dt \right)^{q/2} dP \right]^{1/q} \\
&= \left\| (1 + |F|^{\alpha}) \left(\int_0^{\infty} (D_t F)^2 dt \right)^{1/2} \right\|_q
\end{aligned}$$

Note that $\frac{1}{q} = \frac{1}{p} + \frac{1}{p/\alpha}$ and so if we apply Holders Inequality above we get that

$$[E \|\varphi'(F)DF\|_H^q]^{1/q} \leq \|1 + |F|^{\alpha}\|_{p/\alpha} \left\| \left(\int_0^{\infty} (D_t F)^2 dt \right)^{1/2} \right\|_p$$

which is finite since $F \in \mathbb{D}^{1,p}$. Or equivalently,

$$\begin{aligned}
\|\varphi'(F)DF\|_H^2 &= \|\varphi(F)\|_q^2 \|DF\|_H^2 \\
&\leq \|\varphi(F)\|_{\frac{p}{\alpha}}^2 \|DF\|_p^2 \\
&= \|\varphi(F)\|_{\frac{p}{\alpha}}^2 \int_0^{\infty} \|D_t F\|_p^2 dt
\end{aligned}$$

□

Nick's Question 3.7 Prop 3.3.2 page 54

I have not been able to verify why $\varphi(F) \in \mathbb{D}^{1,q}$. By definition, this means that we need to show that there exists $X_n \in \mathcal{S}$ such that

1. $X_n \rightarrow \varphi(F)$ in $L^q(\Omega)$
2. $D(X_n) \rightarrow D(\varphi(F))$ in $L^q(\Omega, H)$

You have made the problem more complex than what it should be. You only need to check the seminorm $\|\cdot\|_{1,p}$ is finite.

By assumption we already have that there exists $\{F_n\} \subset \mathcal{S}$ such that $F_n \rightarrow F$ in $L^p(\Omega)$ and also $DF_n \rightarrow DF$ in $L^p(\Omega, H)$. We then take a sequence of mollifiers, $\{\alpha_n\}$ and consider the functions $\varphi_k(x) = (\varphi * \alpha_k)(x)$ which by the original assumption on φ are $\alpha + 1$ holder continuous. You cannot say a function is β -Hölder continuous with $\beta \geq 1$. That means the function is a constant.

Now, we notice that by properties of the mollifier since $\varphi(F) \in L^q(\Omega)$ then

$$E|\varphi_k(F_j) - \varphi(F_j)|^q \xrightarrow{j \rightarrow \infty} 0$$

Next, since φ is $\alpha + 1$ holder continuous then we have

$$\begin{aligned}
E|\varphi(F_j) - \varphi(F)|^q &\leq KE|F_j - F|^{q(\alpha+1)} \\
&= KE|F_j - F|^p \rightarrow 0
\end{aligned}$$

and thus we conclude as k and j go to infinity that $\varphi_k(F_j) \rightarrow \varphi(F)$ in $L^q(\Omega)$ and $\varphi_k(F_j) \in \mathcal{S}$.

I am unable to show why $D_t(\varphi_k(F_j)) \rightarrow D_t(\varphi(F))$ in $L^q(\Omega, H)$.

However since $F_j = f^j(B(h_1^j), \dots, B(h_n^j))$, we see that

$$D_t(\varphi_k(F_j)) = \sum_{i=1}^n \frac{\partial}{\partial x_i} (\varphi * \alpha_k)(F_j) h_i^j(t)$$

$$\begin{aligned}
&= \sum_{i=1}^n (\varphi' * \alpha_k)(F_j) \frac{\partial f^j}{\partial x_i} h_i^j(t) \\
&= (\varphi' * \alpha_k)(F_j) \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(t) \\
&= (\varphi' * \alpha_k)(F_j) D_t(F_j).
\end{aligned}$$

We know that $\{\alpha_k\}$ are mollifiers so $(\varphi' * \alpha_k)(F_j) \rightarrow \varphi'(F_j)$ almost surely and by assumption $D_t(F_j) \rightarrow D_t(F)$ in $L^p(\Omega, H)$. But I still am having trouble rigorously showing that $D_t(\varphi_k(F_j)) \rightarrow D_t(\varphi(F_j))$ in $L^p(\Omega, H)$. **You have two indices, you need to take the limit once for just one indices, not the two at the same time. More precisely,**

$$\|D_t(\varphi_k(F_j) - D_t(\varphi(F)))\|_p \leq \|D_t(\varphi_k(F_j)) - D_t(\varphi(F_j))\|_p + \|D_t(\varphi(F_j)) - D_t(\varphi(F))\|_p.$$

Actually, you need to split to three such differences. I will explain to you during the meeting.

Nick's Question 3.8 Exercise 3.4 Let $M = \sum_{t \in [0,1]} X_t$. The hint says to approximate this by taking the sup over a finite set. We know that since X_t is continuous in t then $M = \sup_{t \in \mathbb{A}} X_t$ where $A = \mathbb{Q} \cap [0, 1]$. Let $\{a_n\}$ be an enumeration of $A \setminus \{0, 1\}$ and let $S_k = \{0, a_1, \dots, a_k, 1\}$ then each S_k are finite and $M_k := \sup_{t \in S_k} X_t$ is an **(Note how to type $\mathbb{Q}$, you need curly braces.)** approximation of M since $M_k \rightarrow M$ almost surely. The hint also tells us to use Corollary 4.2.5 but that requires us to find $D_t M_k$ which I am unsure of how to do. The hint says that we need to use exercise 3.3 which is chain rule for Lipschitz functions but I do not know how that applies. This corollary would also require us to show that $M_k \xrightarrow{L^2(\Omega)} M$ which I have been unable to show either. **Try harder... This is an interesting problem that is related the hitting problems that Robert Dalang and Eulalia have done a lot work in the SPDE setting.**

3.2 Exercises

Exercise 3.3 Let α_n be a sequence of mollifiers and $\varphi_k := (\varphi * \alpha_k)$. Since φ is Lipschitz with Lipschitz constant K then this implies that $\varphi'_k \leq K$ and in addition φ' exists almost everywhere and is also bounded by K . By proposition 3.3.2 we have that

$$D_t(\varphi_k(F)) = \varphi'_k(F) D_t(F)$$

Also note that

$$\begin{aligned}
|\varphi_k(F) - \varphi(F)|^2 &= \left| \int_{\mathbb{R}} (\varphi(F - y) - \varphi(F)) \alpha_k(y) dy \right|^2 \\
&\leq K \left| \int_{\mathbb{R}} y \alpha_k(y) dy \right|^2 \\
&\leq K'
\end{aligned}$$

Hence $|\varphi_k(F) - \varphi(F)|^2$ is bounded above and also converges to 0 almost surely so by the dominated convergence theorem we have that

$$\mathbb{E}|\varphi_k(F) - \varphi(F)|^2 \rightarrow 0$$

and so the hypothesis of corollary 4.2.5 is satisfied. This implies that $\varphi(F) \in \mathbb{D}^{1,2}$ and

$$D_t(\varphi_n(F)) \rightarrow D_t(\varphi(F))$$

weakly in $L^2(\Omega, H)$. Lastly, since $\varphi'_k(F)D_t(F)$ is bounded in $L^2(\Omega, H)$ and also

$$\varphi'_k(F)D_t(F) \rightarrow GD_t(F)$$

almost surely where G is a random variable bounded by K then

$$\varphi'_k(F)D_t(F) \rightarrow GD_t(F)$$

weakly in $L^2(\Omega, H)$ as well. Since weak limits are unique then we must have

$$D_t(\varphi(F)) = GD_t(F)$$

However note that in general we have as a property of mollifiers that $\varphi'_n(x) \rightarrow \varphi'(x)$ almost surely. But we said before that $\varphi'(x)$ only exists almost surely since φ is Lipschitz. Therefore if the law of F is absolutely continuous with respect to Lebesgue measure then we have $G = \varphi'(F)$ almost surely. Indeed, let $\mathcal{N}$ be the null set where φ' is undefined. If the law of F is absolutely continuous with respect to Lebesgue measure then $P(F \in \mathcal{N}) = 0$ and we must have $G = \varphi'(F)$ almost surely.

Exercise 3.4 I found a variant of this problem [here](#) where they consider $\{B_t\}_{t \in [0,1]}$ instead of an arbitrary centered Gaussian processes $\{X_t; t \in [0,1]\}$. They use the idea I was trying, which is we want to consider the function $f(x) = \max_{i \in \{1, \dots, n\}} x_i$ for $x \in \mathbb{R}^n$. f is a Lipschitz function and if we consider $\{q_n\}$ an enumeration of $\mathbb{Q} \cap [0,1]$ and consider $Y_n = (X_{q_1}, \dots, X_{q_n})$ then $M_n := \max_{i \in \{1, \dots, n\}} \{X_{q_1}, \dots, X_{q_n}\} = f(Y_n)$ and $M_n \rightarrow M$ almost surely. However in the above linked pdf they claim without proof that that $M_n \rightarrow M$ in $L^2(\Omega)$ which I have so far been unable to show.

I was unaware that there is an extension of the chain rule for Lipschitz functions that is given in the above linked pdf that says we may consider Lipschitz functions on $\mathbb{R}^d$ and random vectors $F = (F_1, \dots, F_n)$ whose components are in $\mathbb{D}^{1,2}$ and then $f(F)$ will also be in $\mathbb{D}^{1,2}$. We apply this above to see that $M_n := \max_{i \in \{1, \dots, n\}} \{X_{q_1}, \dots, X_{q_n}\} = f(Y_n) \in \mathbb{D}^{1,2}$. Next we define the sets

$$A_1 = \{M_n = X_{q_1}\}$$

...

$$A_k = \{M_n \neq X_{q_1}, M_n \neq X_{q_2}, \dots, M_n = X_{q_k}\}, 2 \leq k \leq n$$

and by doing so

$$\begin{aligned} DM_n &= D \left(\sum_{k=1}^n 1_{A_k} X_{q_k} \right) \\ &= \sum_{k=1}^n 1_{A_k} D(X_{q_k}) \end{aligned}$$

where the last equality is to follow because Lemma 3.4.2 on page 56 but I need to verify why. Lets suppose for now that $\{X_t\}$ are Brownian motions. Then

$$D(X_{q_k}) = 1_{[0, q_k]}(t) \tag{3.10} \quad \boxed{\text{brown}}$$

and the above becomes

$$\begin{aligned} DM_n &= \sum_{k=1}^n 1_{A_k} D(X_{q_k}) \\ &= \sum_{k=1}^n 1_{A_k} 1_{[0, q_k]}(t) \end{aligned}$$

and we have that

$$\begin{aligned}
\mathbb{E} \|DM_n\|_H^2 &= \mathbb{E} \left\| \sum_{k=1}^n 1_{A_k} 1_{[0, q_k]}(t) \right\|_H^2 \\
&= \mathbb{E} \int \left(\sum_{k=1}^n 1_{A_k} 1_{[0, q_k]}(t) \right)^2 dt \\
&= \sum_{k=1}^n \mathbb{E} \left(1_{A_k} \int 1_{[0, q_k]}(t) dt \right)
\end{aligned}$$

where the last inequality follows from the disjointness of the A_k . Continuing the calculations we see

$$\begin{aligned}
\mathbb{E} \|DM_n\|_H^2 &= \sum_{k=1}^n \mathbb{E} \left(1_{A_k} \int 1_{[0, q_k]}(t) dt \right) \\
&= \sum_{k=1}^n q_k \mathbb{E} (1_{A_k}) \\
&\leq \sum_{k=1}^n \mathbb{E} (1_{A_k}) \\
&\leq \mathbb{E} (\Omega) = 1
\end{aligned}$$

and so we apply Corollary 4.2.5 to conclude that $M = \sup_{t \in [0, 1]} X_t \in \mathbb{D}^{1,2}$ and that $DM_n \rightarrow DM$ in the weak topology of $L^2(\Omega, H)$ which means that for all $G \in \mathbb{D}^{1,2}$

$$\langle DM_n, G \rangle_{L^2(\Omega, H)} \rightarrow \langle DM, G \rangle_{L^2(\Omega, H)}$$

To prove that $DX = 1_{[0, T]}$ we note that $\sum_{k=1}^n 1_{A_k} q_k$ which converges almost surely to T . Then we can conclude that $DM_n = \sum_{k=1}^n 1_{A_k} 1_{[0, q_k]}(t)$ converges to $1_{[0, T]}$ almost surely. Since DM_n converges both weakly and almost surely then a result from functional analysis says that these limits must be equal and we see that $DM = 1_{[0, T]}$.

Now if we go back to (3.10) and assume that $\{X_t\}$ is a centered Gaussian process then I am unsure of how to continue. For example we can not calculate $D(X_t)$ for any $t \in [0, 1]$ unless we know that $X_t = \int_0^\infty h(s) dB_s$ for some $h \in H$ but I do not think that this can be claimed.